

Northeastern University

Report for Experiment #3

Motion in One and Two Dimensions

Arefin Mahmood

February 28, 2025

Abstract

This experiment analyzed the motion of a hover puck in one and two dimensions using video tracking software. The objective was to study displacement, velocity, and acceleration, and to confirm that motion in one direction is independent of motion in the perpendicular direction. The experiment involved recording and analyzing the puck's movement on an inclined plane. Experimental results were compared with theoretical predictions, and sources of error were identified.

Introduction

According to Newtonian mechanics, an object in motion follows specific kinematic equations that describe its position, velocity, and acceleration. This experiment aimed to investigate these principles by analyzing the motion of a puck moving on an inclined air table. Galileo's early experiments on motion provided the theoretical foundation, demonstrating that motion in perpendicular directions remains independent.

The objectives of this experiment were:

To measure displacement, velocity, and acceleration in one and two dimensions.

To confirm that horizontal and vertical components of motion are independent.

To compare experimental acceleration with theoretical values.

The experiment was conducted using a glass air table, a hover puck, and video tracking software to record motion data. The collected data was analyzed using velocity-time graphs and acceleration calculations.

Investigation 1

Setup & Procedure

The glass air table was leveled, and a hover puck was placed at its center. The video tracking software was calibrated by marking the four corners of the table and adjusting the color filtering settings to track the puck. Data on the puck's position was recorded for 10 seconds while it remained stationary.

Data & Analysis

The x and y positions determine the measurement uncertainty. The standard deviation of position measurements was calculated to estimate random error, which increased during motion due to motion blur.

t(s)	x1(cm)	y1(cm)	x2(cm)	y2(cm)	
0.0333	32.622	32.287	0	65	
0.0667	32.637	32.267	0	65	
0.1	32.641	32.265	0	65	
0.1333	32.636	32.268	0	65	
0.1667	32.64	32.279	0	65	
0.2	32.635	32.272	0	65	
0.2333	32.631	32.265	0	65	
0.2667	32.64	32.28	0	65	
0.3	32.638	32.274	0	65	
0.3333	32.634	32.28	0	65	
0.3666	32.631	32.284	0	65	
0.4	32.634	32.271	0	65	
0.4333	32.641	32.273	0	65	
0.4666	32.64	32.268	0	65	
0.5	32.64	32.27	0	65	
0.5333	32.634	32.269	0	65	
0.5667	32.635	32.271	0	65	
0.6	32.644	32.265	0	65	
0.6333	32.636	32.268	0	65	
0.6667	32.643	32.287	0	65	
0.7	32.631	32.267	0	65	
0.7333	32.639	32.272	0	65	
0.7667	32.636	32.276	0	65	
0.8	32.635	32.268	0	65	
0.8333	32.636	32.265	0	65	
0.8666	32.643	32.275	0	65	
0.9	32.638	32.282	0	65	
0.9333	32.64	32.266	0	65	

Investigation 2

Setup & Procedure

The puck was placed at the top of the inclined plane and released without an initial push. It moved downward, bouncing off the bottom edge. The motion was recorded, and position vs. time data was extracted.

Data & Analysis

The velocity in the y-direction was computed using: $v_y = \frac{y_2 - y_1}{t_2 - t_1}$ A velocity vs. time graph was plotted, and its slope was used to determine acceleration. Two straight-line regions were fitted with trendlines to extract acceleration values.

Results & Discussion

The calculated acceleration was compared with theoretical predictions. Errors due to friction, table leveling inaccuracies, and motion blur were discussed. The acceleration due to gravity was estimated based on the slope of the velocity-time graph.

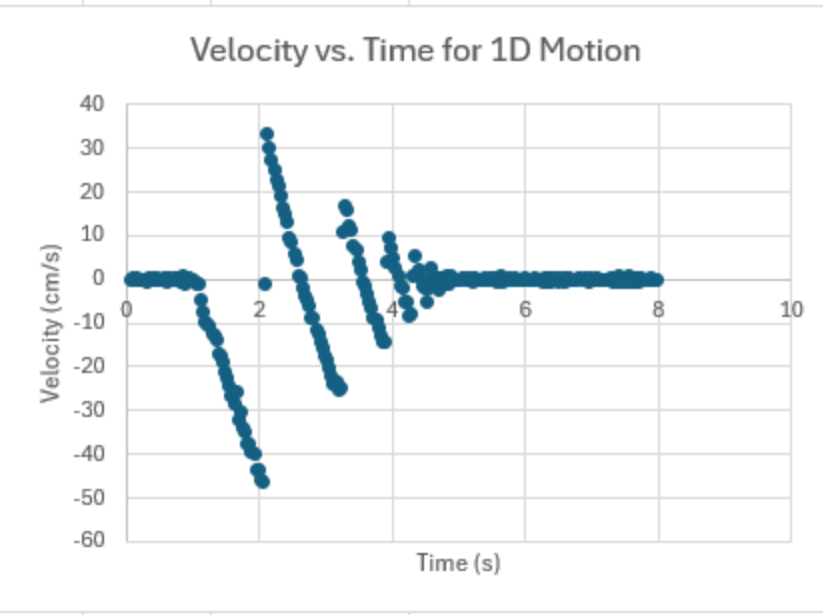


Table 1: Velocity Data for 1D Motion

J	K	L	M	N	O	P	Q
t(s)	x1(cm)	y1(cm)	x2(cm)	y2(cm)	Vy	tavg	erro Vy
0.0333	32.357	31.783	0	65	-0.179640719	0.05	1.497006
0.0667	32.361	31.777	0	65	0.18018018	0.08335	1.501502
0.1	32.362	31.783	0	65	-0.24024024	0.11665	1.501502
0.1333	32.366	31.775	0	65	0.239520958	0.15	1.497006
0.1667	32.359	31.783	0	65	0.06006006	0.18335	1.501502
0.2	32.352	31.785	0	65	0.09009009	0.21665	1.501502
0.2333	32.353	31.788	0	65	0.089820359	0.25	1.497006
0.2667	32.35	31.791	0	65	-0.15015015	0.28335	1.501502
0.3	32.362	31.786	0	65	-0.36036036	0.31665	1.501502
0.3333	32.36	31.774	0	65	0.119760479	0.35	1.497006
0.3667	32.354	31.778	0	65	0.27027027	0.38335	1.501502
0.4	32.356	31.787	0	65	-0.09009009	0.41665	1.501502
0.4333	32.354	31.784	0	65	0.359281437	0.45	1.497006
0.4667	32.352	31.796	0	65	-0.24024024	0.48335	1.501502
0.5	32.362	31.788	0	65	-0.03003003	0.51665	1.501502
0.5333	32.347	31.787	0	65	-0.269461078	0.55	1.497006
0.5667	32.37	31.778	0	65	0.39039039	0.58335	1.501502
0.6	32.352	31.791	0	65	-0.39039039	0.61665	1.501502
0.6333	32.357	31.778	0	65	0.359281437	0.65	1.497006
0.6667	32.356	31.79	0	65	-0.27027027	0.68335	1.501502
0.7	32.361	31.781	0	65	0.33033033	0.71665	1.501502
0.7333	32.353	31.792	0	65	-0.209580838	0.75	1.497006
0.7667	32.358	31.785	0	65	0.36036036	0.78335	1.501502

Investigation 3

Setup & Procedure

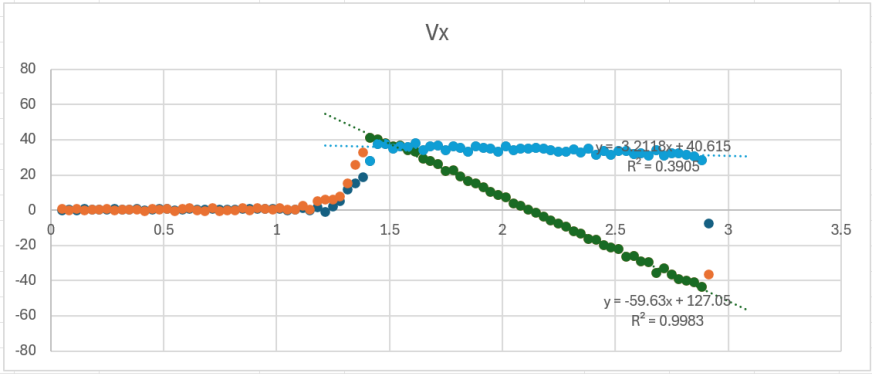
The puck was launched from the lower-left corner of the inclined table, following a parabolic trajectory. The motion was recorded and analyzed.

Data & Analysis

The components of velocity were computed as follows: $v_x = \frac{x_2 - x_1}{t_2 - t_1}$, $v_y = \frac{y_2 - y_1}{t_2 - t_1}$ Both velocity components were plotted against time, and trendlines were used to extract acceleration values.

Results & Discussion

The independence of horizontal and vertical motions was confirmed. The acceleration in the y-direction matched the expected value, while the x-direction remained nearly constant, as predicted by theoretical models.



y=-59. 6299x+1275 y=-3.2122x+40.61

Table 2: Velocity Data for 2D Motion

t(s)	x1(cm)	y1(cm)	x2(cm)	y2(cm)	Tavg	Vx	erro Vx	Vy	erro Vy
0.0333	6.351	6.006	0	65	0.05	-0.3293	0.01497	0.41916	0.01497
0.0667	6.34	6.02	0	65	0.08335	0.12012	0.01502	-0.3604	0.01502
0.1	6.344	6.008	0	65	0.11665	-0.2703	0.01502	0.48048	0.01502
0.1333	6.335	6.024	0	65	0.15	0.50898	0.01497	-0.2695	0.01497
0.1667	6.352	6.015	0	65	0.18335	0.09009	0.01502	0.09009	0.01502
0.2	6.355	6.018	0	65	0.21665	0.03003	0.01502	-0.0601	0.01502
0.2333	6.356	6.016	0	65	0.25	0.20958	0.01497	0.53892	0.01497
0.2667	6.363	6.034	0	65	0.28335	0.36036	0.01502	-0.4805	0.01502
0.3	6.375	6.018	0	65	0.31665	0.03003	0.01502	-0.0901	0.01502
0.3333	6.376	6.015	0	65	0.35	0.02994	0.01497	0	0.01497
0.3667	6.377	6.015	0	65	0.38335	0.39039	0.01502	0.06006	0.01502
0.4	6.39	6.017	0	65	0.41665	-0.1802	0.01502	-0.8408	0.01502
0.4333	6.384	5.989	0	65	0.45	0.11976	0.01497	0.68862	0.01497
0.4667	6.388	6.012	0	65	0.48335	0.45045	0.01502	0.3003	0.01502
0.5	6.403	6.022	0	65	0.51665	0.78078	0.01502	0.54054	0.01502
0.5333	6.429	6.04	0	65	0.55	-0.1796	0.01497	-0.8383	0.01497
0.5667	6.423	6.012	0	65	0.58335	0.27027	0.01502	0.36036	0.01502
0.6	6.432	6.024	0	65	0.61665	0.51051	0.01502	0.96096	0.01502
0.6333	6.449	6.056	0	65	0.65	0.26946	0.01497	-0.3892	0.01497
0.6667	6.458	6.043	0	65	0.68335	0.15015	0.01502	-0.8108	0.01502
0.7	6.463	6.016	0	65	0.71665	0.42042	0.01502	0.84084	0.01502
0.7333	6.477	6.044	0	65	0.75	0.05988	0.01497	-0.6886	0.01497
0.7667	6.479	6.021	0	65	0.78335	0.09009	0.01502	-0.1502	0.01502
0.8	6.482	6.016	0	65	0.81665	0.3003	0.01502	-0.2703	0.01502
0.8333	6.492	6.007	0	65	0.85	0.47904	0.01497	1.01796	0.01497
0.8667	6.508	6.041	0	65	0.88335	0.57057	0.01502	-0.4204	0.01502
0.9	6.527	6.027	0	65	0.91665	0.39039	0.01502	1.02102	0.01502

Investigation 4

Analysis

The measured accelerations from Investigations 2 and 3 were averaged to estimate the acceleration due to gravity. The theoretical value was 9.81 m/s².

Results & Discussion

The experimental value was compared with the accepted value. Discrepancies were attributed to systematic errors such as air resistance and measurement limitations.

Tavg	Vx	Vy	a	g(cm/s^2)
0.05	-0.32934	0.419162	59.6299	12.50304
0.08335	0.12012	-0.36036	3.2122	0.673526
0.11665	-0.27027	0.48048	54.33	11.39177
0.15	0.508982	-0.26946	53.591	11.23682
0.18335	0.09009	0.09009		8.951292
0.21665	0.03003	-0.06006		
0.25	0.209581	0.538922		
0.28335	0.36036	-0.48048		
0.31665	0.03003	-0.09009		
0.35	0.02994	0		
0.38335	0.39039	0.06006		
0.41665	-0.18018	-0.84084		
0.45	0.11976	0.688623		
0.48335	0.45045	0.3003		
0.51665	0.780781	0.540541		
0.55	-0.17964	-0.83832		
0.58335	0.27027	0.36036		
0.61665	0.510511	0.960961		
0.65	0.269461	-0.38922		
0.68335	0.15015	-0.81081		
0.71665	0.42042	0.840841		
0.75	0.05988	-0.68862		
0.78335	0.09009	-0.15015		

Conclusion

This experiment successfully demonstrated fundamental principles of motion. The independence of motion in perpendicular directions was validated. Measured acceleration values were close to theoretical predictions, with minor deviations due to systematic errors.

The primary sources of error included motion blur, air resistance, and potential inaccuracies in the leveling of the table. Future improvements could include using a higher frame-rate video tracker and more precise table leveling.

Questions

Time to reach the bottom – I found that the times were similar but differed slightly due to experimental errors. Possible reasons for discrepancies include friction, table leveling imperfections, or minor inconsistencies in the puck's release mechanism.

Acceleration on an incline – I derived the acceleration using $a_\theta = g \sin\theta$, where is g gravitational acceleration and θ is the angle of incline. As it increases, the acceleration component along the incline increases, leading to faster motion.

Intercept of velocity plot – The intercept represents the initial velocity at release. If the intercept is nonzero, it suggests that I unintentionally applied an initial force during release. Experimental errors might also have contributed to small deviations from the expected zero intercept.

Effect of setting $\theta = 0$: if $\theta = 0$, the incline would be flat, meaning no component of gravitational force would act along the surface. The puck would remain at rest unless I applied an external force.

Required launch angle – In Investigation 3, I needed to launch the puck so that it reached the opposite corner of the square table while traveling equal distances in x and y. Using projectile motion equations, I determined that the optimal launch angle was 45 degrees.